PORTFOLIO

Lian Cataytay

Data Analyst & Economist

Decision-ready analytics, dashboards, and research workflows built with an economist's lens.

Lian turns raw datasets into clear, explainable outputs for business, policy, and people decisions. His work spans econometrics, business intelligence, machine learning, customer analytics, credit risk, workforce intelligence, and applied research.

Based in the Philippines

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13
portfolio projects

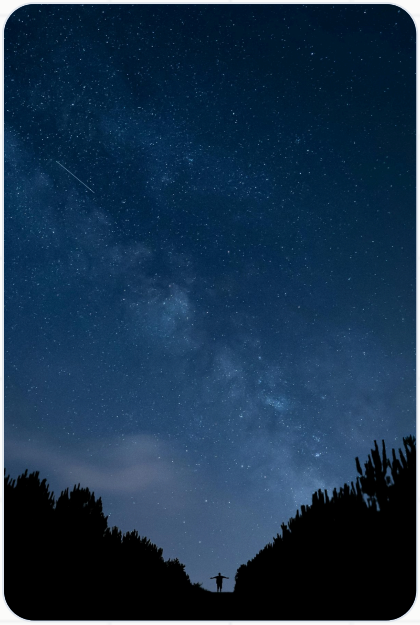
BI
Power BI and Tableau dashboards

ML
classification and explainability

ECON
policy and econometric analysis

PROFILE

Analytics for better decisions



Lian is a data analyst and economist from the Philippines. His work spans econometrics, business intelligence, and machine learning. He has built projects involving retention analytics, AI-powered churn prediction, banking risk analysis, and workforce intelligence.

Data is only valuable when it leads to better decisions. Many organizations already have the numbers; what they lack is a clear way to interpret them. That is the gap he focuses on closing.

Right now, he is especially interested in research and AI, with a focus on making analytics practical, explainable, and useful to decision-makers.

01

Business Intelligence

Dashboard-ready KPIs, stakeholder reporting, performance views, and decision-focused metric design.

02

Econometrics & Research

Time-series analysis, ARDL/ECM modeling, correlation analysis, diagnostics, and policy-facing summaries.

03

Machine Learning

Classification analytics, churn-risk scoring, explainability with SHAP, feature importance, and model interpretation.

04

Data Storytelling

Clean analysis narratives that connect methods, findings, business implications, and next steps.

Python · Pandas · Power BI · Tableau · Stata · R · Excel · SHAP

FEATURED WORK

Customer and retention intelligence

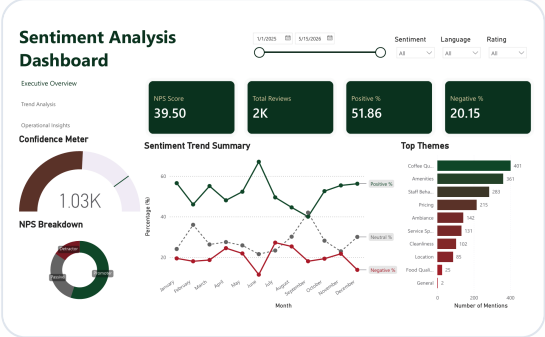
NLP · SENTIMENT ANALYSIS · BUSINESS INTELLIGENCE

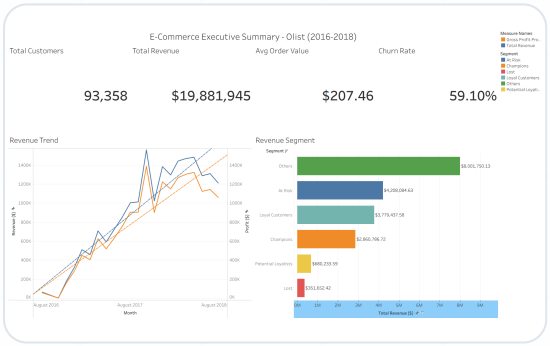
Customer Sentiment Intelligence Dashboard

Sentiment analysis of customer reviews and business-facing metrics for interpreting customer pain points and satisfaction trends.

TOOLS Python, Power BI, Pandas, NLP workflow

FINDING Of 500 reviews, 41.6% were positive and 25.4% negative (NPS 30); coffee quality, staff behavior, and amenities were the top themes.





CUSTOMER ANALYTICS · RETENTION · SEGMENTATION

E-commerce Customer Retention Analytics

Customer behavior analysis focused on retention, segmentation, cohort patterns, and churn-risk insights.

TOOLS Python, Pandas, Tableau, Cohort analysis

FINDING Across 93,358 customers and $19.9M in revenue, churn was 59.1%; Champions were only 6.9% of customers while 14.0% were at risk.

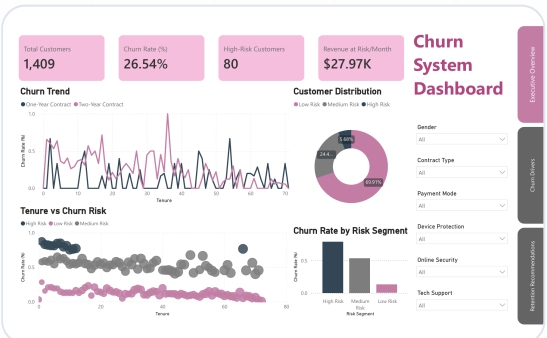
MACHINE LEARNING · EXPLAINABILITY · CHURN RISK

AI Customer Churn Intelligence System

An explainable churn-prediction workflow using machine learning, SHAP, and retention-focused reporting.

TOOLS Python, scikit-learn, SHAP, Power BI

FINDING The High Risk segment churned at 78.8% versus 12.9% for Low Risk; contract type, pricing, and service usage drove churn.



RESEARCH & POLICY

Productivity and labor market analysis

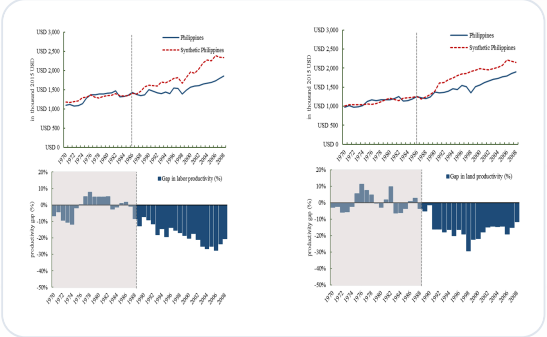
AGRICULTURAL ECONOMICS · SYNTHETIC CONTROL METHOD · IMPACT EVALUATION

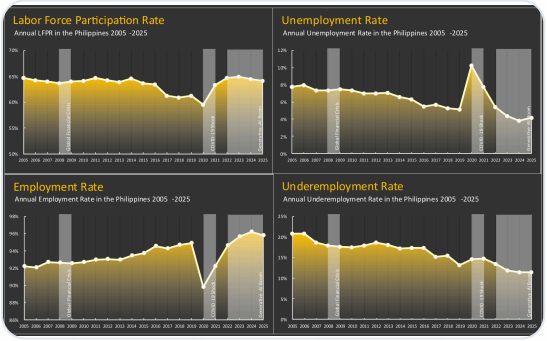
CARP and Agricultural Productivity: A Synthetic Control Study

A synthetic control evaluation of how the 1988 Comprehensive Agrarian Reform Program affected Philippine agricultural productivity.

TOOLS R, R Markdown, Synthetic control, TFP analysis

FINDING From 1989 to 2008, labor productivity ran about 18.2% below the synthetic control, land productivity 16.1%, and TFP about 9%.





LABOR ECONOMICS · TIME-SERIES TRENDS · POLICY INSIGHT

Philippines Labor Market Trend Analysis

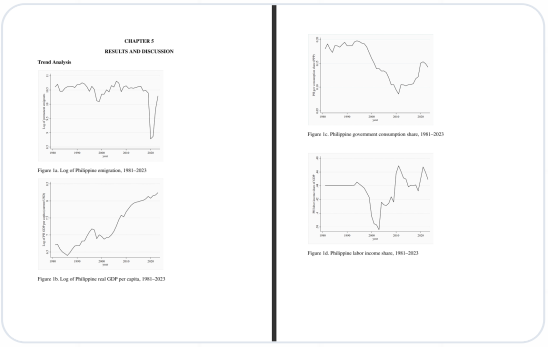
A labor-market analysis of employment quality, workforce composition, and underemployment patterns in the Philippines.

TOOLS Python, Pandas, Time-series analysis, Excel

FINDING In 2025, participation was 64.1%, unemployment 4.2%, and underemployment 11.4%, almost three times the unemployment rate.

APPLIED RESEARCH & RISK

Macroeconomics, risk, and media analysis



MACROECONOMICS · MIGRATION · ARDL MODELING

Philippine Emigration ARDL Macroeconomic Analysis

An econometric analysis of how macroeconomic conditions relate to Philippine emigration trends.

TOOLS Stata, ARDL bounds testing, ECM modeling, Time-series diagnostics

FINDING Five of seven ARDL models were cointegrated; a 1% rise in PH GDP per capita is linked to a 0.46% fall in long-run emigration.

MEDIA INSIGHTS · THEMATIC ANALYSIS · POLICY RISK

Media Insight Report: Philippine Inflation and Policy Debate

A media insight report on the 2026 inflation spike, built from 15 media and official sources for a retail client persona.

- Coded coverage into cost-of-living, wage, and social protection themes
- Mapped wage-push, pricing, and reputational risks and opportunities

BANKING ANALYTICS · CREDIT RISK · FINANCIAL PERFORMANCE

Banking Risk and Financial Performance Analysis

A financial-risk project identifying borrower and loan characteristics associated with default exposure.

0.72 ROC AUC

53% default rate in the top risk band vs 2% in the lowest

HR ANALYTICS · ATTRITION RISK · WORKFORCE INTELLIGENCE

Employee Attrition Risk and Workforce Intelligence Analysis

A workforce intelligence project identifying employee groups and teams with stronger attrition-risk patterns.

Risk Hotspots

Team-level retention signals

FORECASTING & OPERATIONS

Public health, finance, and marketing analytics

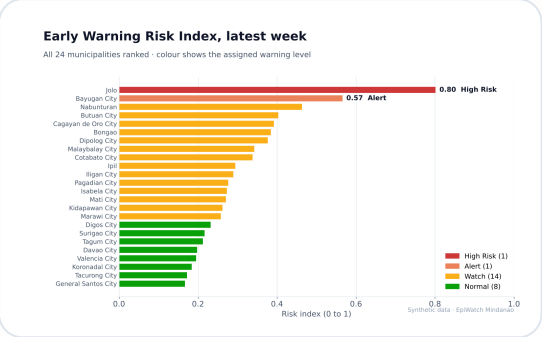
PUBLIC HEALTH ANALYTICS · FORECASTING · RISK SCORING

EpiWatch Mindanao Public Health Risk Forecasting

A simulated early warning system with case forecasting, anomaly detection, and a composite risk index for 24 municipalities.

TOOLS Python, Pandas, NumPy, Excel, Power BI-ready tables

FINDING 3,024 municipality-week records; the latest week had 1 High Risk, 1 Alert, 14 Watch, and 8 Normal municipalities.



FP&A · FORECASTING · VARIANCE ANALYSIS

FP&A Revenue, Cost & Expense Forecasting Dashboard

Monthly finance reporting with budget variance, scenario forecasts, KPI trends, and anomaly review.

98.2% accuracy

Forecast accuracy on $281.6M revenue

MARKETING ANALYTICS · ADS · BUDGET ALLOCATION

eCommerce Marketplace Ads Performance Dashboard

Ads reporting across four marketplaces with ACoS/TACoS KPIs, slow-moving SKUs, and budget recommendations.

6.80 ROAS

$86.7M ad revenue; 485 slow-moving SKUs flagged

CRM · DATA QUALITY · EVENT MARKETING

CRM & Event Marketing Data Hygiene System

CRM cleanup with duplicate resolution, contact validation, and event follow-up workflows.

1,187 contacts

313 duplicates removed; 469 invalid emails flagged

Available for data analysis, business intelligence, research, and AI-assisted analytics work.

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